Calculus

Final

Aug/18/2026 – Sep/16/2026 lecture

Partial derivatives

Aug.18, 2026 lecture

Partial derivative, which implied by ∂(Del) is considered to apply derivation to single and multi-variable function.

For multi-variable function f(x_1, x_2, …, x_i), partial derivative is its derivative with respect to one of those variables, with the others held constant.

Definition

Define that the partial derivative with respect to x_1 in multi-variable function f as

$$\frac{\partial}{\partial x_1} f(x_1, x_2, \dots, x_i) = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \dots, x_i) - f(x_1, x_2, \dots, x_i)}{h}$$

In addition, we can define total differential df as

$$df = \sum_i^N \frac{\partial f}{\partial x_i} dx_i$$

In case of differentiate function partially for the one variable, for instance x, we say: “take the partial differentiation respect to x”.

For example, let’s find partial differentiation of function $f(x, y) = x^2 + y^2$ respect to x.

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial x} x^2 + \frac{\partial}{\partial x} y^2 = 2x + 0$$

When differentiate functions partially, other variables are treated as constant. So, partial differentiation of variables unrelated to x is same as differentiation of constant (=0).
That said, when differentiating the product of two variables, the other variable does not disappear.

For example, partial differentiation of function $f(x,y) = xy$ respect to x is

$$\frac{\partial}{\partial x} f(x,y) = \frac{\partial}{\partial x} xy = y$$

It also applied to exponential, logarithmic, fractional functions.

Skills

Partial differentiation respect to one variable x for the function consisting of two variables, another variable must be treated as constant as below,

$$\frac{\partial}{\partial x}\{f(x) + f(y)\} = \frac{d}{dx} f(x) + 0$$
$$\frac{\partial}{\partial x}\{f(x) \cdot f(y)\} = f(y) \cdot \frac{d}{dx} f(x)$$

Also, Partial differentiation for single variable function is same as its differentiation.

$$\frac{\partial}{\partial x} f(x) = \frac{d}{dx} f(x)$$

In addition, Partial differentiation works the same way for higher-order derivatives. And it is denoted by $f_{xx\ldots}$ or $\frac{\partial^n}{\partial x^n} f$.

Partially differentiated function also can be partially differentiated against with same variable ether another variable. And it doesn't depend on the order of differentiation.

For example, for the function $f(x,y) = x^2 + y^2$,

$$f_{xy} = \frac{\partial^2}{\partial y \partial x}(x^2 + y^2) = \frac{\partial}{\partial y} 2x = 0$$

$$f_{yx} = \frac{\partial^2}{\partial x \partial y}(x^2 + y^2) = \frac{\partial}{\partial x} 2y = 0$$

They yield the same value regardless of the order of differentiation, and this principle applies to all continuously differentiable functions.

Example 1:

Find the derivative of given implicit function with respect to x via using partial derivatives.

$$xy + x\sin xy - \frac{1}{y} = x^2$$

Sol)

Take partial differentiation respect to x,

$$\frac{\partial}{\partial x}\left(xy + x\sin xy - \frac{1}{y}\right) = \frac{\partial}{\partial x}x^2$$

$$right\ side \rightarrow \frac{\partial}{\partial x}xy + \frac{\partial}{\partial x}x\sin xy - \frac{\partial}{\partial x}\frac{1}{y} = y + \sin xy + xy\cos xy + 0$$

$$left\ side \rightarrow \frac{\partial}{\partial x}x^2 = \frac{d}{dx}x^2 = 2x$$

$$\therefore\ y + \sin xy + xy\cos xy = 2x$$

Example 2:

Find the derivative of given function respect to each variable (Part of revision resources A6)

$$f(x,y) = 4e^{6x}\sin 3y$$

Sol)

Take partial differentiation respect to x,

$$\frac{\partial}{\partial x}(4e^{6x}\sin 3y) = 24e^{6x}\sin 3y$$

Also respect to y,

$$\frac{\partial}{\partial y}(4e^{6x}\sin 3y) = 12e^{6x}\cos 3y$$

$$\therefore\ 24e^{6x}\sin 3y\ /\ 12e^{6x}\cos 3y$$

Differentiation of multi-variable functions

Sep.07, 2026 lecture
In this case, we only address two variable (3rd dimensional) functions.

Image the function z consists of two variables x, y such as below,

$$z(x, y) = x^2 + y^2$$

If one of the variables depends on another variable, it can be differentiated by usual operator. However, in this case, that is not the case.

To find stationary point of function having two variables doesn't related to each other, partial derivative must be used.

Function has stationary points at x_0, y_0 value satisfies condition below,

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial y} f(x, y) = 0$$

Although the ordered pairs found here are stationary points, it cannot be assumed whether they are local maxima, minima, or saddle points without further analysis. To classify these stationary points, we need to apply the Second Derivative Test using the **Hessian matrix**

First, construct the Hessian matrix H of second-order partial derivatives at the stationary point (x_0, y_0),

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$$

Next, compute the determinant D of the Hessian matrix:

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = (f_{xx} \cdot f_{xy}) - {f_{xy}}^2$$

Finally, classify the point based on the sign of Determinants and f_{xx}

(We haven't learnt matrix yet, just memories $D = (f_{xx} \cdot f_{xy}) - {f_{xy}}^2$)

Hessian matrix

To classify these stationary points, we need to apply the Second Derivative Test using the Hessian matrix

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$$

And its determinant,

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = (f_{xx} \cdot f_{xy}) - {f_{xy}}^2$$

Stational points can be classified as follows table below,

Conditions	Classification
$D > 0 \; and \; f_{xx} > 0$	Local minimum
$D > 0 \; and \; f_{xx} < 0$	Local maximum
$D < 0$	Saddle point
$D = 0$	Inconclusive

Example 1:

List the stationary point of function below:

$$z = x^2 + y^2$$

Sol)

$$\frac{\partial z}{\partial x} = 2x \, , \frac{\partial^2 z}{\partial x^2} = 2 \; and \; \frac{\partial z}{\partial y} = 2y \, , \frac{\partial^2 z}{\partial y^2} = 2 \; and \; \frac{\partial z}{\partial x \partial y} = 0$$

$$function \; have \; stationary \; point \; only \; at \; x_0 = \; 0, y_0 = 0$$

$$By \; Hessian \; matrix, D = 2 \cdot 2 - 0 = 4 > 0 \; and \; f_{xx} > 0 \rightarrow local \; minimum$$

$$\therefore (0,0), local \; minimum$$

Introduction to integration

Aug.22, 2026 lecture
Concept of integration and trapezoidal rule.

Integration was designed to find “area” of function.

To find precise area enclosed by function, break down the function into smaller pieces, until its length is negligible. And the function will be much like straight line.
Then the area of small pieces is approximately similar to the area of rectangular (or trapezoidal) box.

$$A = base \times height \approx \Delta x \times f(x)$$

The Trapezoidal rule is a numerical method used to approximate the value of a definite integral or the area under a curve by dividing the region into connected trapezoids.

$$A \approx \frac{1}{2}\frac{b-a}{n}[f(a) + 2\{\sum_{i=1}^{n-1} f(a + \frac{i(b-a)}{n})\} + f(b)]$$

To explain any further, we’d have to get into things we haven't learned yet, and I don't really want to bring that up again.

Anti-derivatives

Aug.18, 2026 lecture
To find out what was the original form of the derivative!

Anti-derivative is differential function F whose derivatives are equal to the original function f(x). The processing of solving anti-derivatives is called “Integration”

Definition

Antiderivative F is defined with original function f as below,

$$F'(x) = f(x)$$

We can also express the anti-derivatives with “integration” symbol as below, which is looks like stretched “S” the first letter of word “Summation”.

While $f(x) = \frac{d}{dx}F(x)$,

$$f(x)dx = dF(x)$$

$$\therefore \int f(x)dx = \int dF(x) = F(x)$$

But it’s not complete definition. let’s make sure about principle of differentiation to find what’s wrong with above form… For any function of the form: a part dependent on x plus a constant,

$$\{f(x) + 1\}' = f'(x)$$

$$\{f(x) + 100\}' = f'(x)$$

$$\{f(x) + 13898269308 \dots\}' = f'(x)$$

…

When we differentiate a function, constant becomes zero. So, when we go backward - which is integration, we know there was *some* constant there, but we don't know what it was. We need the constant C because differentiation destroys information about constants.

Definition

Antiderivative F is defined with original function f as below,

$$\int f(x)dx = F(x) + C$$

C, called “Integration constant”.

Basic skills for finding anti-derivatives

There’re several skills to find anti-derivatives of functions.

Techniques

$$\int x^n dx = \frac{1}{n+1}x^{n+1} + C \qquad \int \frac{1}{x}dx = \ln|x| + C$$

$$\int e^x dx = e^x + C \qquad \int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x \; dx = -\cos x + C \qquad \int \cos x \; dx = \sin x + C$$

$$\int (\sec x)^2 \; dx = \tan x + C \qquad \int (\csc x)^2 \; dx = -\cot x + C$$

$$\int \sec x \tan x \; dx = \sec x + C \qquad \int \csc x \cot x \; dx = -\csc x + C$$

Example 1:

Prove that the anti-derivative of sum of constant and e^x itself.

Sol)

$$use\ definition\ of\ e, \qquad e = \lim_{x\to 0}[1+x]^{\frac{1}{x}}$$

$$\frac{d}{dx}e^x = \lim_{h\to 0}\frac{e^{x+h}-e^x}{h} = e^x \lim_{h\to 0}\frac{e^h-1}{h}$$

$$let\ e^h - 1 = t, then\ h = \ln t + 1$$

$$e^x \lim_{h\to 0}\frac{e^h-1}{h} = e^x \lim_{t\to 0}\frac{t}{\ln(t+1)} = e^x \lim_{t\to 0}\frac{1}{\frac{\ln(t+1)}{t}} = e^x \lim_{t\to 0}\frac{1}{\ln\left[(t+1)^{\frac{1}{t}}\right]}$$

$$= e^x \frac{1}{\ln\left[\lim_{t\to 0}(t+1)^{\frac{1}{t}}\right]} = e^x \frac{1}{\ln e} = e^x$$

$$while\ \frac{d}{dx}e^x = e^x, \int de^x = \int e^x dx = e^x + c$$

Example 2:

For anti-derivative F(x) of given function, F (0) = 0. Find the anti-derivative of given function.

$$f(x) = 5x^4$$

Hint) Do we have to add integration constant?

Sol)

$$\int 5x^4 dx = x^5 + c$$

$$F(0) = 0 \rightarrow 0^5 + c = 0\ thus\ c = 0$$

$$\therefore x^5$$

Substitution

Aug.19, 2026 lecture
It's easiest part of integration!!!!!

Substitution used for finding the anti-derivatives of complex functions.
Let's look some examples.

Some functions couldn't be integrated directly. For instance,

$$f(x) = \frac{1}{\sqrt{6 - x^2}}$$

We can't evaluate the anti-derivative of this directly.
In another case, there's a function

$$f(x) = \frac{1}{1 + e^x}$$

WTF.

Both looks completely not integrable, but it integrated by substitute new variables corresponding to current variables.

Substitution – For single variable

Let's look at the function below,

$$f(x) = \frac{1}{1 + x}$$

And anti-derivatives of function must be:

$$F(x) = \int \frac{1}{1 + x} dx$$

1+x is in denominator of fraction, it couldn't integrate directly by x.
If we simply define 1+x to new variable, like below,

$$let\ u = 1 + x \rightarrow \frac{dx}{du} = 1$$

dx becomes du, then we can re-write

$$F(x) = \int \frac{1}{1 + x} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|1 + x| + C$$

Techniques – u-method

To integrate complex fractional functions have form like below,

$$\frac{x^{n-1}}{a+x^n}$$

$$let\ a+x^n=u \rightarrow \frac{du}{dx}= nx^{n-1}$$

Substitution – Function has a sqrt

Now look at the function that has root. It's one of the simplest.

Let's get a function that has square root like below:

$$f(x) = \sqrt{1+x}$$

And anti-derivatives of function must be:

$$F(x) = \int \sqrt{1+x}\, dx$$

1+x is in denominator of fraction, it couldn't integrate directly by x.
If we simply define 1+x to new variable, like below,

$$let\ u = 1 + x \rightarrow \frac{dx}{du} = 1$$

dx becomes du, then we can re-write

$$F(x) = \int \sqrt{1+x}\, dx = \int \sqrt{u} du = \frac{2}{3}u^{\frac{3}{2}} + C = \frac{2}{3}(1+x)^{\frac{3}{2}} + C$$

Techniques – Root function

To integrate complex root functions have form like below,

$$x^{n-1}\sqrt[i]{a+x^n}$$

$$let\ a + x^n = u \rightarrow \frac{du}{dx} = nx^{n-1}$$

Substitution – Trigonometric substitution

That's the trickiest one. (Also, the funny one)
Image that the functions have both above condition instantly.

Like the below,

$$f(x) = \frac{1}{\sqrt{4 - x^2}}$$

If letting $4+x^2 = u$,

$$\frac{du}{dx} = -2x \rightarrow dx = -\frac{du}{2x}$$

$$\int \frac{1}{\sqrt{4 - x^2}} dx = -\int \frac{1}{\sqrt{u}} \cdot \frac{du}{2x}$$

The extra x couldn't be canceled out!
Thus, to eliminate x, we need to account for other variables. Trigonometric functions are good instances for alternative.

Let x to some trigonometric function. Like below:

$$let\ x = 2\sin\theta$$

$$\frac{dx}{d\theta} = 2\cos\theta \rightarrow dx = 2\cos\theta\, d\theta$$

$$\therefore \int \frac{1}{\sqrt{4 - x^2}} dx = 2\int \frac{\cos\theta}{\sqrt{4 - (\sin\theta)^2}} d\theta = 2\int \frac{\cos\theta}{2\cos\theta} d\theta = \theta + C$$

$$x = 2\sin\theta \rightarrow \theta = \sin^{-1}\frac{x}{2}$$

$$Thus, \int \frac{1}{\sqrt{4 - x^2}} dx = \sin^{-1}\frac{x}{2}$$

It is simply expressed as an inverse function of sine.

Also, we able to use tangent, and secant for antiderivatives of complicated functions.

Techniques – Trigonometric substitution

To integrate complex functions, consisting of one of them below,

* Even without root signs (radicals), trigonometric substitution is highly effective whenever you encounter expressions that match or can be converted into the standard Pythagorean identity forms

$$\sqrt{a^2 - x^2} \rightarrow let\ x = a\sin x\,, at\ the\ range\ of\ -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\sqrt{a^2 + x^2} \rightarrow let\ x = a\tan x\,, at\ the\ range\ of\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$\sqrt{x^2 - a^2} \rightarrow let\ x = a\sec x\,, at\ the\ range\ of\ 0 \leq \theta < \frac{\pi}{2}\ or\ \pi \leq \theta < \frac{3\pi}{2}$$

Integration of multiplied functions

Aug.20, 2026 lecture
fuck

Due to integration is antidifferentiation (i.e. opposite direction of differentiation), integrating each multiplied function separately and then multiplying them couldn't work at all.

$$\int x\ln x\,dx \neq \int xdx \cdot \int \ln x\,dx$$

Then, how do we integrate that?
Assume that the Integrand is product of two functions, and one of them is derivatives of some function. Then, by the product rules,

$$\frac{d}{dx}f(x)\cdot g(x) = \frac{df(x)}{dx}\cdot g(x) + f(x)\cdot\frac{dg(x)}{dx}$$

$$\rightarrow\ f(x)\cdot\frac{dg(x)}{dx} = \frac{d}{dx}f(x)\cdot g(x) - \frac{df(x)}{dx}\cdot g(x)$$

$$\therefore \int\left\{f(x)\cdot\frac{dg(x)}{dx}\right\}dx = f(x)\cdot g(x) - \int\left\{\frac{df(x)}{dx}\cdot g(x)\right\}dx$$

Or simply,

$$\int f(x) \cdot g'(x)dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x)dx$$

Via using above relation, integration of multiple functions is simply calculated.

$$\int x \ln x \, dx = \int \left(\frac{x^2}{2}\right)' \ln x \, dx = \frac{x^2}{2}\ln x - \frac{1}{2}\int x^2 \cdot (\ln x)' \, dx$$

During derivative of ln(x) = 1/x, second term of right side simply calculated as integration of x.

Techniques – Integration by parts

For multiplied function,

$$f(x) \cdot g'(x)$$

Its anti-derivative will be

$$\int f(x) \cdot g'(x)dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x)dx$$

Example 1:

Find anti-derivatives of given function:

$$f(x) = xe^x$$

Sol)

$$let\ x = f(x), e^x = g'(x)$$

$$\int xe^x dx = xe^x - \int e^x dx$$

$$Thus, = xe^x - \int e^x dx = e^x(x-1) + c$$

Tip: Remember "LIATE". *It's best to picked your u and dv in this order.*

→ *Logarithmic, Inverse trigonometric, Algebric, Trigonometric, Exponential*

$$\therefore e^x(x-1) + c$$

Integration of fraction

Fractional form is also hard to integrate. Some kind of functions can be solved by substitution, but most of functions are like below,

$$\int \frac{1}{x^2+2x-3}dx$$

If function have nth degree polynomial in its denominator, we can express that to sum of two or more fractions.

$$\frac{1}{x^2+2x-3} = \frac{A}{(x+3)} + \frac{B}{(x-1)}$$

$$B(x+3) + A(x-1) = x^2+2x-3 \rightarrow A = \frac{1}{4}, B = -\frac{1}{4}$$

$$\therefore \frac{1}{x^2+2x-3} = \frac{1}{4(x+3)} - \frac{1}{4(x-1)}$$

$$\rightarrow \int \left\{\frac{1}{4(x+3)} - \frac{1}{4(x-1)}\right\}dx = \frac{1}{4}\ln\left|\frac{x+3}{x-3}\right| + C$$

Techniques – partial fraction

$$\frac{c}{AB} = \frac{c}{B-A}\left(\frac{1}{A} - \frac{1}{B}\right)(while\ c = constant)$$

$$\frac{N(x)}{(ax+b)(cx+d)} \equiv \frac{A}{(ax+b)} + \frac{B}{(cx+d)}$$

$$\frac{N(x)}{(ax+b)^2} \equiv \frac{A}{(ax+b)} + \frac{B}{(ax+b)^2}$$

$$\frac{N(x)}{(ax+b)(x^2+c)} \equiv \frac{A}{(ax+b)} + \frac{Bx+c}{(x^2+c)}$$

(sorry)

Integration of complex Trigonometric functions

You can imagen that the differentiation of trigonometric functions.
Consider the integration is calculation have opposite directions to differentiation, we can easily find the result.

For example, from differentiation of sin(x),

$$\frac{d}{dx}\sin x = \cos x$$

$$d\sin x = \cos x \; dx$$

$$\sin x + c = \int \cos x \; dx$$

We can recognize that the integration of cos(x) is sin (x).
Integration of sin(x) is also found easily via using differentiation of cos(x).
How about product of trigonometric function?

$$\int \cos 3x \sin 4x \, dx$$

To Integrate multiplied functions, it must be separated as the sum of two or more functions.
Via using relationship below,

$$\cos A \sin B = \frac{1}{2}\{\sin(A+B) - \sin(A-B)\}$$

$$\int \cos 3x \sin 4x \, dx = \frac{1}{2}\int \{\sin(7x) - \sin(-x)\} \, dx$$

It simply expressed to sum of two functions.

Techniques – Angle addition and subtraction formulas

For trigonometric function,

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$$

$$\sin\alpha\sin\beta = -\frac{1}{2}[\cos(\alpha+\beta) - \cos(\alpha-\beta)]$$

***Useful but not on the lecture**

Trigonometric – reduction formular

Techniques – integration of $\sin^n x$ on $x \geq 2$

$$\int (\sin x)^n \, dx = -\frac{1}{n}(\sin x)^{n-1} \cos x + \frac{(n-1)}{n} \int (\sin x)^{n-2} \, dx \; (n \geq 2)$$

Pf.

$$let \; I_n = \int (\sin x)^n \, dx$$

$$\int (\sin x)^n \, dx = \int \sin x \cdot (\sin x)^{n-1} \, dx$$

$$via \; using \int f(x) \cdot g'(x) \, dx = f(x)g(x) + \int f'(x) \cdot g(x) \, dx$$

$$\rightarrow \int \sin x \cdot (\sin x)^{n-1} \, dx = -(\sin x)^{n-1} \cos x + (n-1) \int (\sin x)^{n-2} (\cos x)^2 \, dx$$

$$the \; second \; term: (n-1) \int (\sin x)^{n-2} (\cos x)^2 \, dx$$

$$= (n-1) \left\{ \int (\sin x)^{n-2} \, dx - \int (\sin x)^n \, dx \right\}$$

$$= (n-1) \left\{ \int (\sin x)^{n-2} \, dx - I_n \right\}$$

$$\therefore \; nI_n = -(\sin x)^{n-1} \cos x + (n-1) \int (\sin x)^{n-2} \, dx,$$

$$I_n = -\frac{1}{n}(\sin x)^{n-1} \cos x + \frac{(n-1)}{n} \int (\sin x)^{n-2} \, dx \; (n \geq 2)$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = (\sin x)^3$$

Techniques – integration of $\cos^n x$ on $x \geq 2$

$$\int (\cos x)^n \, dx = \frac{1}{n}(\cos x)^{n-1} \sin x + \frac{(n-1)}{n} \int (\cos x)^{n-2} \, dx \; (n \geq 2)$$

Pf.

$$let \; I_n = \int (\cos x)^n \, dx$$

$$\int (\cos x)^n \, dx = \int \cos x \cdot (\cos x)^{n-1} \, dx$$

$$via \; using \int f(x) \cdot g'(x) \, dx = f(x)g(x) + \int f'(x) \cdot g(x) \, dx$$

$$\rightarrow \int \cos x \cdot (\cos x)^{n-1} \, dx = (\cos x)^{n-1} \sin x + (n-1) \int (\cos x)^{n-2} (\sin x)^2 \, dx$$

$$the \; second \; term: (n-1) \int (\cos x)^{n-2} (\sin x)^2 \, dx$$

$$= (n-1) \left\{ \int (\cos x)^{n-2} \, dx - \int (\cos x)^n \, dx \right\}$$

$$= (n-1) \left\{ \int (\cos x)^{n-2} \, dx - I_n \right\}$$

$$\therefore \; nI_n = (\cos x)^{n-1} \sin x + (n-1) \int (\cos x)^{n-2} \, dx,$$

$$I_n = \frac{1}{n}(\cos x)^{n-1} \sin x + \frac{(n-1)}{n} \int (\cos x)^{n-2} \, dx \; (n \geq 2)$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = (\cos x)^3$$

Area of function

Aug.25, 2026 lecture
fuck

The purpose of integration is not only to find its anti-derivative, but also to find area enclosed with the function. Usually, bounded integration is used for area. Without a range, area might diverge to infinity.

$$\int_a^b f(x)dx$$

And bound also separated centred at $c \in [a, b]$

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

* Too much time taken to prove this rule.

For instance, area enclosed by $3x^2$ from 0 to 3 is

$$3\int_0^3 x^2dx = [x^3]_0^3 = 27$$

And area of function must be positive and couldn't be subtracted to each area. Therefore, we must put an absolute value on all integration, and we need to split the interval of integration to account for the regions where the function is negative.

Sine and cosine function are great examples of it. Normal integration of sine is 0 as below.

$$\int_0^{2\pi} \sin x\, dx = 0$$

So, we must separate function considering the negative parts of it for evaluate "Area"

$$\int_0^{2\pi} \sin x\, dx = \int_0^{\frac{\pi}{2}} \sin x\, dx - \int_{\frac{\pi}{2}}^{\pi} \sin x\, dx + \int_{\pi}^{\frac{3\pi}{2}} \sin x\, dx - \int_{\pi}^{\frac{3\pi}{2}} \sin x\, dx = 4$$

However, we must face horrible things than above.

Area enclosed by two functions

Boundary also the function. Bounded to a to b, it means also function surrounded by x = 0 (x axis), y = a and y = b. So, Boundary can change to another function.

How can we calculate the area of the larger circle, excluding the part covered by the smaller circle? It's easy. Just substrate smaller one from bigger one.

$$A = A_{larger} - A_{smaller}$$

This concept holds true for integration too. Just substrate smaller one from bigger one. For example, area enclosed by x^2 and x from 0 to 1.

$$x^2 < x$$

So, the area will be

$$A = \int_0^1 x - x^2 \, dx$$

Techniques – Area

For the two functions,

$$\forall x \in [a, b], f(x) > g(x)$$

Its area:

$$A = \int_a^b f(x) - g(x) dx$$

However, if the range extended to [0, 2], x^2 larger than x after 1.
In this case, separate range centred at increase point to [0,1] and [1,2]. Integrate separately.

$$A = \int_0^1 x - x^2 \, dx + \int_1^2 x^2 - x \, dx$$

Techniques – Area

If,

$$\exists c \in [a,b]\,, \exists \delta > 0, \forall x \in [a,b],$$
$$[(c-\delta < x < c \to f(x) > g(x)) \wedge (c < x < c+\delta \to f(x) < g(x))]$$

Its area:

$$A = \int_a^c f(x) - g(x)dx + \int_c^d g(x) - f(x)dx$$

Length of the function

Length of the function describes as below:

$$S = \int_a^b \sqrt{1 + \left(\frac{d}{dx}f(x)\right)^2}\, dx$$

Derivation of above formular is not that difficult. For the infinitesimal length ds,

$$ds = \sqrt{dx^2 + dy^2}$$

While dx and dy also the infinitesimal length of x, and y components.
For integration, put dx out of the square root,

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \frac{dy}{dx^2}^2}\, dx = \sqrt{1 + \left(\frac{d}{dx}f(x)\right)^2}\, dx$$

$\frac{dy}{dx^2}^2$ is not second-order differentiation.

So, the total length of the function got from integrating ds.

Differential equations

Sep.10 - 15, 2026 lecture
Almost there! Last but least! I hate this shiiiii!

Newton's 3rd law looks like this.

$$F = ma$$

Consider the acceleration is second-order differentiation of the displacement, equation above expressed bit different.

$$F = m\frac{d^2x}{dt^2}$$

Differential equations are one of the equations have differentiation. Some kinds of equations easily solved by integration (most of them are not but).

Differential equations – linear first order

If the equations are linear and have form of below:

$$\frac{dy}{dx} = f(x) \cdot f(y)$$

Solution can find via integration after moving dx and dy to each side

$$\frac{1}{f(y)}dy = f(x)dx$$

Most solutions to these types of equations take the form of an exponential function.

Techniques – Saparation

$$\frac{dy}{dx} = f(x) \cdot f(y) \rightarrow \frac{1}{f(y)}dy = f(x)dx$$

If the functions have form below:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Rewriting the right-hand side as the derivative of a product leads to the previous form, which is the easiest form to solve.
Multiply unknown function I(x) to both sides then

$$I(x)\frac{dy}{dx} + I(x)P(x)y = I(x)Q(x)$$

What we want is…

$$I(x)\frac{dy}{dx} + I(x)P(x)y = \frac{d}{dx}\,[I(x)\cdot y] = \; y\frac{d}{dx}I(x) + I(x)\frac{dy}{dx}$$

Thus,

$$\frac{d}{dx}I(x) = \; I(x)P(x)$$

$$\frac{1}{I(x)}dI(x) = P(x)dx \rightarrow \ln|I(x)| + c = \int P(x)dx$$

Integration constant c can be any number so just let that to 0, then

$$I(x) = \exp(\int P(x)dx)$$

While exp(x) = e^x. And we call I(x) to "Integration factor"

Techniques – Saparation

To solve equation form of

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Use integration factor below,

$$I(x) = \exp(\int P(x)dx)$$

Makes it to the terms of derivative of a product

$$\frac{d}{dx}\,[I(x)\cdot y] = \; I(x)Q(x)$$

Differential equations – linear second order

In general, second-order linear differential equations have form of below.

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x)$$

If R(x) = 0, homologous, R(x)≠ 0, non-homogeneous.

And it describes as its "characteristic equations".

Assume that the homologous equations have a solution, $y = e^{mx}$. Then,

$$y' = me^{mx}$$

$$y'' = m^2e^{mx}$$

Substitute to above equation then it organized to terms of m,

$$m^2e^{mx} + P(x)me^{mx} + Q(x)e^{mx} = 0$$

Divide both sides by e^{mx} than we can get its characteristic equations in form of quadratic equations.

$$m^2 + P(x)m + Q(x) = 0 \; or \; am^2 + bm + c = 0$$

And the solution of the differential equations determined by it's determinants.

Solutions of homologous second-order differential equations

While $D = b^2 - 4ac$

Conditions	Classification
$D > 0$	$y = Ae^{m1x} + Be^{m2x}$
$D = 0$	$y = (Ax + B)e^{mx}$
$D < 0$ (y = a±bi)	$y = e^{ax}(A\cos bx + B\sin bx)$

If the equation is not homogeneous,

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x)$$

First, let R(x) = 0, find general solution like above.

And then, substitute corresponding forms of solution depending on the form of R(x)

Finally, add both solutions to finalise.

Method of Undetermined Coefficients

While $y = y_h + y_p$, particular solutions will be

$Polynomial$	$y = Ax^n + Bx^{n-1} + \cdots$
$Exponential$	$y = Ae^{nx}$
$Trigonometric$	$Corresponding\ Trigonometric$

하시발끝